

HANNAH-MARIE LAMLE

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Miami, FL - 33167, USA

RESEARCH INTERESTS

trait-based ecological modeling, coral macro-algal competition, reefscape ecology

EDUCATION

- **Florida International University** August 2023 - May 2028
P.h.D. Biological Sciences
◦ Advisor: Dr. Rolando Santos
◦ Dissertation: "Quantifying colorful communities: Community assembly, function and resilience of corals throughout South Florida using spectral traits" Miami, FL, USA
- **Indiana University** August 2018 - May 2022
B.S. Environmental Science
◦ Certification: Underwater Resource Management Bloomington, IN, USA
◦ Honors Thesis: "Coral Reefs under Threat: Could Repopulating the West Indian Spider Crab Help?"

RESEARCH EXPERIENCE

- **Graduate Research Assistant** [🌐] August 2023 - Present
Florida International University Santos Seascape Ecology Lab
◦ Conducting coral bleaching experiment using heat stress to understand relationship of coral reflectance to spectral traits Miami, FL, USA
◦ Developing novel technology to implement underwater hyperspectral imaging to traditional photogrammetry methods
- **Research Associate** [🌐] August 2022 - July 2023
Indiana University Center for Underwater Science
◦ Field monitoring of IUCN recognized endangered coral species on shipwreck sites in Dominican Republic Bloomington, IN, USA
◦ Management of biological resource conservation of archaeological sites
- **Research Assistant** [🌐] August 2021 - May 2022
Indiana University Geobiology of Invertebrates Lab
◦ Curated datasets with 20+ years of coral abundance data across 6 archaeological sites in the Florida Keys and Dominican Republic Bloomington, IN, USA
- **Research Assistant** [🌐] August 2020 - May 2022
Indiana University Center for Underwater Science
◦ Assisted with archaeological artifact research and maintenance (conservation, historical data collection, field excavation and maintenance) Bloomington, IN, USA
◦ Created government reports and interpretive materials for underwater archaeological sites
- **Intern** [🌐] May 2021 - August 2021
Mote Marine Laboratory International Center for Coral Reef Research and Restoration
◦ Responsible for over 500 coral fragments, including day-to-day husbandry activities and preparation of fragments to be out-planted Summerland Key, FL, USA
◦ Assisted with growth experiments using photogrammetry techniques to monitor coral growth
- **Research Assistant** [🌐] August 2020 - May 2021
Indiana University Maxwell Research Lab
◦ Responsible for priming, sampling and measuring of tulip tree core samples to aid in identification of historical climate events Bloomington, IN, USA
- **Intern** [🌐] October 2018 - May 2019
Indiana University Integrated Program in the Environment
◦ Conducted guided research with a mentor assessing persistence of document preservation centers in the anthropocene Bloomington, IN, USA
◦ Presented research at annual Indiana University sustainability symposium

PROJECTS

- **Monitoring coral reef through a digital eye: Creating and implementing a hyperspectral imaging photomosaic approach to assess sediment-turf conditions and coral health**

Tools: remote sensing (underwater hyperspectral imaging, photogrammetry), Coralnet, Taglab, Viscore, multivariate statistics, R

 - Assisted with project proposal development and writing and creation of conceptual diagrams.
 - Established protocols and methodologies for data collection
 - Obtained necessary permitting and agreements with EPA, FIU, FWC and NOAA.

August 2025 - Present
- **Show me your colors: Using hyperspectral analysis to quantify coral health**

Tools: spectroscopy, laboratory analysis (spectrophotometer, hemocytometer), coral husbandry, R

 - Designed and developed an experimental framework to investigate influence of physiological trait values on coral spectral reflectance profile
 - Built aquaria setup to house coral fragments during experiment
 - Coordinating logistics and preparations for implementation, including coral acquisition, methodological refinements, collaborations.

August 2024 - Present
- **Assess Impacts of Trawling on Species and Habitats within Biscayne Bay**

Tools: Structure-from-Motion Photogrammetry

 - Assisting with field data collection using photogrammetry

September 2025 - Present
- **Incorporating Structure-from-motion Photogrammetry into Sedimentation Monitoring**

Tools: Structure-from-Motion Photogrammetry, benthic cover quantification, habitat complexity quantification (chain and tape method)

 - Assisting with field data collection using photogrammetry, line point intercept, and chain-and-tape rugosity

November 2025 - Present
- **Stream Flow Reduction Experiment (StreamFRE)**

Tools: Animal tagging equipment (PIT tags, elastomer)

 - Conducted bi-annual field sampling for the Luquillo LTER, utilizing PIT tags and colored elastomer to track shrimp movement in mountain step-pools.

February 2024 - July 2025
- **Fish In Seagrass Habitats: Seascape Connectivity Across Protected Ecosystems (FISHSCAPE)**

Tools: Underwater quadrat photography

 - Supported the use of GPS technology and ground-truthing to train machine learning models for mapping seagrass habitats.

January 2024 - July 2025
- **Restoring the three-dimensional structure of hurricane-impacted coral reefs in Puerto Rico**

Tools: Coralnet, TagLab, R

 - Assisted with data analysis tracking coral colony growth and survivorship across multi-year observational study
 - Assisted with data analysis estimating benthic community coverage in large area images created via photogrammetry
 - Assisted with statistical analysis of population modeling

August 2023 - December 2024
- **RAPID: Here we go again - The fate of *Diadema antillarum* and Caribbean reefs in the 21st century**

Tools: Coralnet, TagLab

 - Assisted with data analysis tracking coral colony growth and survivorship across multi-year observational study
 - Assisted with data analysis estimating benthic community coverage in large area images created via photogrammetry

August 2023 - December 2024
- **NSF IRES: Transdisciplinary Coral Reef Science**

Tools: Pacific Coral Identification, benthic percent cover methods, benthic habitat complexity methods

 - Learned laboratory and field Pacific coral identification methods
 - Conducted data collection for size structure of three pacific coral species (*Acropora cf. hyacinthus*, *Acropora cf. nasuta*, *Pocillopora cf. meandrina* for comparison to historical data from Mermaid Reef, Lizard Island
 - Analyzed changes in size-structure and demography of three coral species (listed above) pre- and post-2024 Great Barrier Reef coral bleaching event

May - June 2024
- **Assessing Functional Trait Dynamics of Caribbean Scleractinia Communities in Puerto Rico's Northeast Marine Corridor**

Tools: Data curation, TagLab

 - Assisted with the creation of regional trait database for selected species and traits of interest

January 2024 - May 2024

- Assisted with data processing via use of TagLab to quantify stony coral species percent cover in photogrammetric plots

• Identification of the 16th Century Punta Espada Shipwreck

January 2022 - July 2023

Tools: underwater photogrammetry, magnetometer use, underwater survey methods for archaeology

- Conducted archaeological survey and site assessment, including mapping of site and associated features
- Assisted with field expedition planning and equipment management
- Restored artifacts taken from the site to aid in pursuits of shipwreck discovery

PUBLICATIONS

Published:

- [1] Lamle, H.M., Mucherheide, S.M., Divish, C., Baelz, J.D. (2025). **Biodiversity of coral species in selected Caribbean Regions**. *IU Journal of Undergraduate Research*, 8(1). <https://doi.org/10.14434/iujur.v8i1.36218>

PRESENTATIONS

Talks:

- [1] Lamle, H., Duran, A., James, R., Santos, R. (March 4, 2026). **Reef structural complexity as a predictor of long-sediment laden algal turf distributions in South Florida**. *Benthic Ecology Meeting 2026*, in Virginia Beach, VA, USA
- [1] Lamle, H., Santos, R. (February 7, 2026). **Mapping Coral Physiology in Spectral Space**. *FIU Biosymposium*, in Miami, FL, USA
- [2] Lamle, H., Duran, A., James, R., Santos, R. (November 21, 2025). **Estimating structural drivers for long sediment laden algal turfs across South Florida reefs**. *ReeFLorida Conference*, in Miami, FL, USA
- [3] Bautista, V., Rivas, N., Lamle, H., Nation, C., James, R., Santos, R. (April 12, 2024). **Assessing functional trait dynamics of Caribbean Scleractinia communities in Puerto Rico's Northeast Marine Corridor**. *Benthic Ecology Meeting*, in Charleston, SC, USA

Posters:

- [1] Lamle, H.M., Santos, R.O. (2025). **Show me your colors: Using spectroscopy to quantify coral health**. *FIU Biosymposium* in Miami, FL, USA.
- [2] Lamle, H.M., Santos, R.O. (2025). **Show me your colors: Using spectroscopy to quantify coral health**. *Reef Futures* in Rivera Maya, Mexico.
- [3] Lamle, H.M., Bautista, V., Rivas, N.I., James, W.R., Santos, R.O. (April 13, 2024). **Exploring the inter- and intra-species pattern of spatial aggregation of coral recruits in Culebra, Puerto Rico**. *Benthic Ecology Meeting* in Charleston, SC, USA
- [4] Lamle, H.M., Baelz, J.D., Johnson, C., Beeker, C.D. (January 2, 2023). **Biology of Shipwrecks in the Dominican Republic: How submerged cultural resources facilitate the growth of endangered and threatened coral species**. *Society of Historical Archaeology Conference* in Lisbon, Portugal.
- [5] Lamle, H.M., Baelz, J.D., Johnson, C., Beeker, C.D. (January 3, 2022). **An updated report for the biology of shipwrecks in the Florida Keys National Marine Sanctuary**. *Society of Historical Archaeology Conference* in Philadelphia, PA, USA.
- [6] Lamle, H.M., Morrison, K., Montoya, R.D. (May 2, 2019). **Document centers and environmental sustainability**. *Indiana University Sustainability Symposium* in Bloomington, IN, USA.

FUNDING

Current:

• South Florida Program

August 2025 - August 2028

Environmental Protection Agency, Region 4: \$750,000



- Three year grant funding proposed novel coral monitoring technique utilizing hyperspectral imaging to non-invasively assess coral health
- Assisted with proposal development and writing, creation of conceptual diagrams, and budget planning

• Graduate Student Coastal and Ocean Research Fellowship

August 2024 - August 2025

Florida SeaGrant: \$25,000



- One year fellowship to fund graduate research activities in coastal/marine science

Completed:

• Student Travel Award

December 2024

FIU CREST CACHÉ: \$500

- Travel funding to Reef Futures Conference
- **Student Travel Award** December 2024
FIU Graduate and Professional Student Council: \$700
 - Travel funding to Reef Futures Conference
- **Student Travel Award** March 2024
FIU CREST CACHÉ: \$500
 - Travel funding to 52nd the Benthic Ecology Meeting
- **Blue World/Oceanic Research Group Undergraduate Marine Conservation Scholarship** March 2022
Women Divers Hall of Fame: \$1000
 - Scholarship to fund undergraduate research activities
- **Provost's Scholarship** August 2018 - January 2022
Indiana University Office of Scholarships: \$4500/semester
 - Scholarship to fund undergraduate tuition
- **Scholar's Award** August 2018 - January 2022
Indiana University O'Neill School of Public and Environmental Affairs: \$1500/semester
 - Scholarship to fund undergraduate tuition
- **Sustainability Scholarship** January 2019
Indiana University Integrated Program in the Environment: \$1000
 - Scholarship to fund undergraduate research experience

SKILLS CERTIFICATIONS

SCUBA diving:

- AAUS Scientific Diver (Florida International University)
- Indiana University Scientific Diver
- PADI Open Water Scuba Instructor (#523283)
- PADI O2 Administration Instructor (#523283)
- PADI Enriched Air Diver
- PADI Underwater Photography
- PADI Search and Recovery
- PADI Cavern Diver
- TDI Ice Diver

- DAN First Aid for Hazardous Marine Life Injuries Provider
- DAN CPR and First Aid (BLS)
- DAN Emergency Oxygen for Scuba Diving Injuries (EO2)

Programming & Data Analysis:

- Proficient in R and PRIMER for statistical analysis, modelling, and processing
- Proficient in Microsoft Suite (Word, Excel, PowerPoint)
- Proficient in Adobe Suite (Lightroom, Photoshop, Premier Pro)
- Proficient in html and css for website development

First Aid:

- DAN Neurological Assessment Provider
- DAN Emergency Oxygen Provider

Technology

- ArcGIS Pro
- VISCORE

- TagLab
- CoralNet
- Agisoft Metashape Pro

Field Work:

- Small vessel operations
- FIU Certified Crew Chief
- Florida boating license
- photogrammetry
- coral disease surveys
- underwater archaeological survey
- underwater magnetometer survey

Laboratory Skills:

- microscopy, cell counting via hemocytometer
- spectrophotometry
- coral microfragmentation

TEACHING & WORKSHOPS

- **Workshop Guest Lecturer** March 19, 2025
Florida Coastal Everglades LTER Graduate Student Group
 Florida International University, FL, USA
 - Website Making Workshop Using RStudio, Quarto and Github
- **Graduate Teaching Assistant** August 2023 - December 2025
Florida International University Department of Biological Sciences
 Miami, FL, USA
 - Marine Protected Areas Online; (5 semesters total) Summer 2024 - December 2025
 - General Biology II Lab Section (2 sections/semester); Spring 2024
 - General Biology I Lab Section (2 sections/semester); Fall 2023
- **Workshop Guest Lecturer** May 6, 2025
Florida Coastal Everglades LTER Graduate Student Group
 Florida International University, FL, USA

- Website Making Workshop Using RStudio, Quarto and Github

- **Visiting Lecturer**

Indiana University Academic Diving Program

- PADI Open Water Diving Course (Certified 60+ students)

August 2022 - June 2023

Bloomington, IN, USA

OUTREACH & VOLUNTEERING

Early Career Researcher (ECR) Committee Member

International Coral Reef Society (ICRS) Student and Early Career Chapter (SECC)

January 2025 - Present

Session Convener

Special Session: Using photogrammetry as a tool to monitor benthic ecosystems

August 22, 2026

Benthic Ecology Meeting 2026

Session Convener

Afternoon Presentations Session

February 7, 2026

FIU Biosymposium

Wawasee High School Alumni Spotlight

Interview with Wawasee School Corporation Communications team discussing my career path and advice to current K-12 students.

August 22, 2025

MAST @ FIU Career Day

Representing FIU Marine Biology Program

February 26, 2025

Family Science Night at Citrus Grove K-8 Center

Representing the Santos Seascape Ecology Lab

March 14, 2024

PROFESSIONAL MEMBERSHIPS

- **International Coral Reef Society Student & Early Career Chapter**

January 2025 - Present

- **American Society for Photogrammetry and Remote Sensing (ASPRS), FIU Chapter**

February 2024 - Present

- **Women Divers Hall of Fame, Associate Member**

March 2022 - March 2023

REFERENCES

1. **Dr. Rolando O. Santos**

Assistant Professor, Department of Biological Sciences

Florida International University

Email: rsantosc@fiu.edu

Phone: +1 (305)-348-3095

Major Advisor

2. **Dr. W. Ryan James**

Research Assistant Professor, Department of Biological Sciences

Florida International University

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Committee Member

3. **Dr. Charles Beeker**

Director, Center for Underwater Science

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Project Supervisor